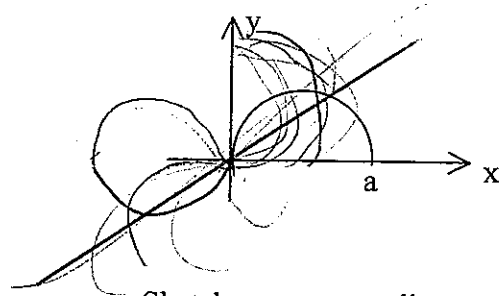


Question 1 10 marks**Marks**

- a) Factorise
- $x^3 + 8$
 - $5x^2 + 5x - 60$
- b) Simplify and express with a rational denominator $\frac{1}{2+\sqrt{3}} - \frac{1}{\sqrt{2}+1}$
- c) Find in simplest exact form, values of x for which $2x^2 - 4x - 7 = 0$
- d) Solve for x and graph on a number line: $|2x-1| > 5$

Question 2 10 Marks**Start this Question on a new page.**

- a) Sketch the region $y > \sqrt{x-1}$
- b) Show that $\frac{x-2}{y} + \frac{y+4}{x} = 0$ is the equation of a circle with centre $(1, -2)$ and radius $\sqrt{5}$.
- c) If the graph of $y = f(x)$ is as follows;



- Sketch on separate diagrams
- $f^{-1}(x)$
 - $f(-x)$

- d) Solve for x : $2x^2 + 5x - 3 \leq 0$

Question 3 10 MarksStart this Question on a new page.**Marks**

- a) Show that $\frac{x+2}{x-1} = 1 + \frac{3}{x-1}$. Hence or otherwise sketch $y = \frac{x+2}{x-1}$. 3
- b) i) Sketch $y = \log_2 x$ 3
ii) State whether it is a function or relation and give reasons.
iii) State the domain.
- c) Solve for x : $\frac{4x}{x-1} \geq 1$. 4

Question 4 10 MarksStart this Question on a new page.

- a) For the function $y = \frac{1}{x^2 - 1}$: 6
- i) State the vertical asymptotes.
- ii) By considering $\lim_{x \rightarrow \pm\infty} \frac{1}{x^2 - 1}$ determine any horizontal asymptotes.
- iii) Show that $y = \frac{1}{x^2 - 1}$ is an even function.
- iv) Find any intercepts.
- v) Graph $y = \frac{1}{x^2 - 1}$, showing these features.
- b) Sketch, indicating any asymptotes and intercepts $y = \frac{|x-1|}{x}$ 4

End of paper.

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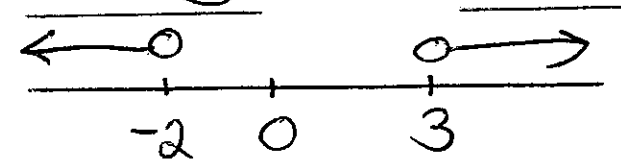
① a) $x^3 + 8 = (x+2)(x^2 - 2x + 4)$ SUM OF 2 CUBES.

(ii) $5x^2 + 5x - 60 = 5(x^2 + x - 12)$
 $= 5(x+4)(x-3)$

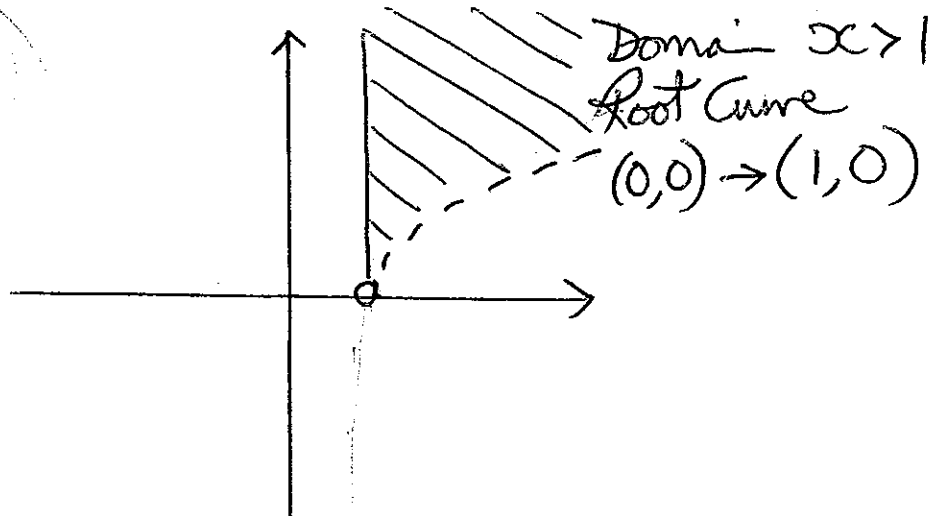
b) $\frac{1}{2+\sqrt{3}} - \frac{1}{\sqrt{2}+1} = \frac{2-\sqrt{3}}{4-3} - \frac{\sqrt{2}-1}{2-1}$
 $= 2-\sqrt{3} - (\sqrt{2}-1) = 3-\sqrt{3}-\sqrt{2}$

c) $2x^2 - 4x - 7 = 0$, $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$
 $x = \frac{4 \pm \sqrt{16 + 56}}{4} = \frac{4 \pm \sqrt{72}}{4} = \frac{4 \pm 2\sqrt{18}}{4} = \frac{4 \pm 6\sqrt{2}}{4}$
 $= 1 \pm \frac{3\sqrt{2}}{2}$

d) $|2x-1| > 5 \therefore 2x-1 > 5 \text{ or } 2x-1 < -5$
 $2x > 6 \quad 2x < -4$
 $x > 3 \quad \text{or} \quad x < -2$



② a) $y > \sqrt{x-1}$



$$b) \frac{x-2}{y} + \frac{y+4}{x} = 0$$

$$\begin{pmatrix} x & x \\ x & y \end{pmatrix}$$

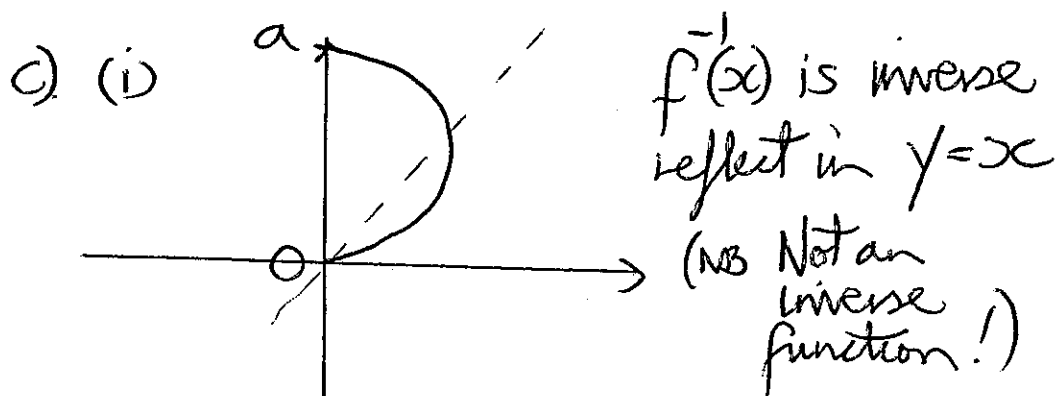
$$x(x-2) + y(y+4) = 0$$

$$x^2 - 2x + y^2 + 4y = 0$$

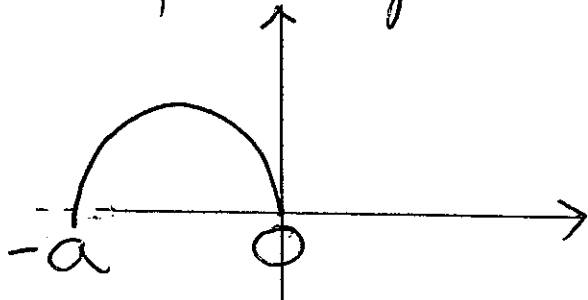
$$(x-1)^2 - 1 + (y+2)^2 - 4 = 0$$

$$(x-1)^2 + (y+2)^2 = 5$$

$\therefore$ Circle centre $(1, -2)$ radius $\sqrt{5}$.



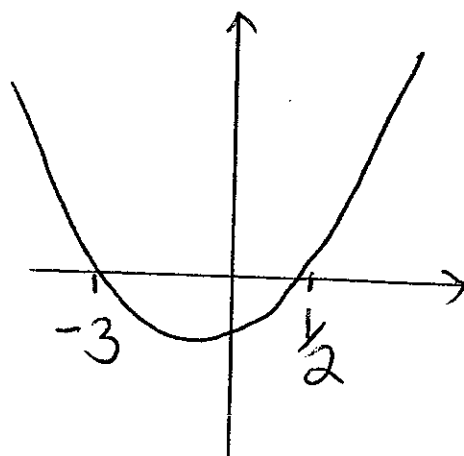
(ii) $f(-x)$ reflect in y axis $x \rightarrow -x$



$$d) 2x^2 + 5x - 3 \leq 0$$

$$(2x-1)(x+3) \leq 0$$

$$\therefore -3 \leq x \leq \frac{1}{2}$$



$$\textcircled{3} \text{ LHS } \frac{x+2}{x-1} = \frac{x-1+3}{x-1} = \frac{x-1}{x-1} + \frac{3}{x-1} = 1 + \frac{3}{x-1} = \text{RHS}$$

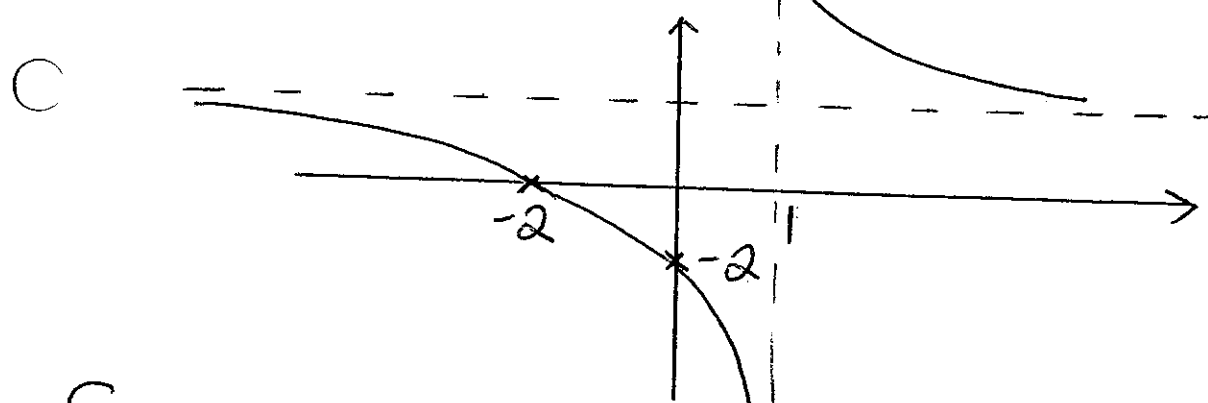
$$y = \frac{x+2}{x-1} = 1 + \frac{3}{x-1}$$

Domain $x \in \mathbb{R} \quad x \neq 1$.

Intercepts $x=0 \quad y = \frac{2}{-1} = -2 \quad (0, -2)$

$y=0 \quad x+2=0 \quad x=-2 \quad (-2, 0)$

As $x \rightarrow \pm \infty \quad \frac{3}{x-1} \rightarrow 0 \quad \therefore y \rightarrow 1$.



{ Alternatively $y-1 = \frac{3}{x-1}$

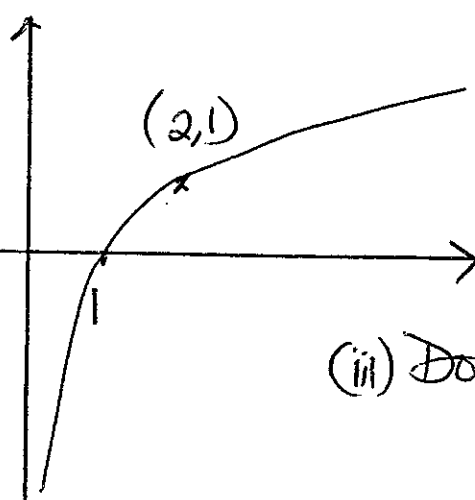
$y = \frac{1}{x}$ Rect Hyp.

$y = \frac{3}{x}$ Stretch $\parallel x$

$y = \frac{3}{x-1} \quad (0,0) \rightarrow (1,0)$

$y-1 = \frac{3}{x-1} \quad (1,0) \rightarrow (1,1)$

b) (i) $y = \log_2 x$



(ii) This is a function since it passes a vertical line test. Each x value maps to a unique y value

(iii) Domain $x \in \mathbb{R} \quad x > 0$

$$c) \frac{4x}{x-1} \geq 1 \quad (x \neq 0)$$

$$x(x-1)^2 \times (x-1)^2$$

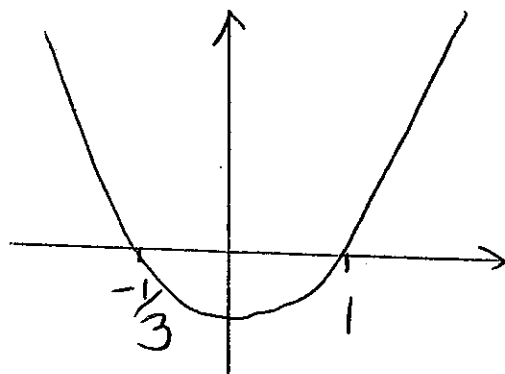
$$4x(x-1) \geq (x-1)^2$$

$$4x^2 - 4x \geq x^2 - 2x + 1$$

$$3x^2 - 2x - 1 \geq 0$$

$$(3x+1)(x-1) \geq 0$$

$$\underline{x \leq -\frac{1}{3} \text{ or } x \geq 1 \text{ But } x \neq 1 \therefore \underline{x > 1}}$$



$$④ a) y = \frac{1}{x^2 - 1} = \frac{1}{(x+1)(x-1)}$$

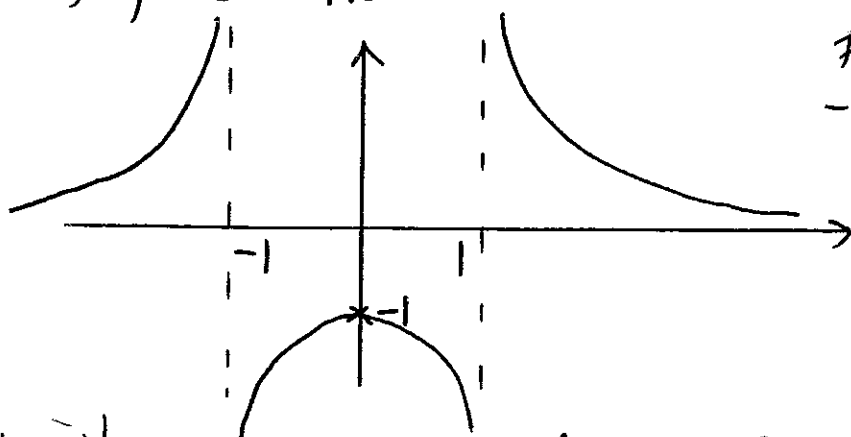
i) Vertical Asymptotes $\Rightarrow$ Denominator Zero $x = \pm 1$ $\circ$

ii) $\lim_{x \rightarrow \pm \infty} \left(\frac{1}{x^2 - 1} \right) = 0 \therefore$ Horizontal Asymptote $y = 0$

iii) $f(x) = \frac{1}{x^2 - 1}$ $f(-x) = \frac{1}{(-x)^2 - 1} = \frac{1}{x^2 - 1} = f(x) \therefore$ Even Fn

iv) y-int, $x = 0$ $y = \frac{1}{-1} = -1$ $(0, -1)$
x-int, $y = 0$ Not Possible

v)

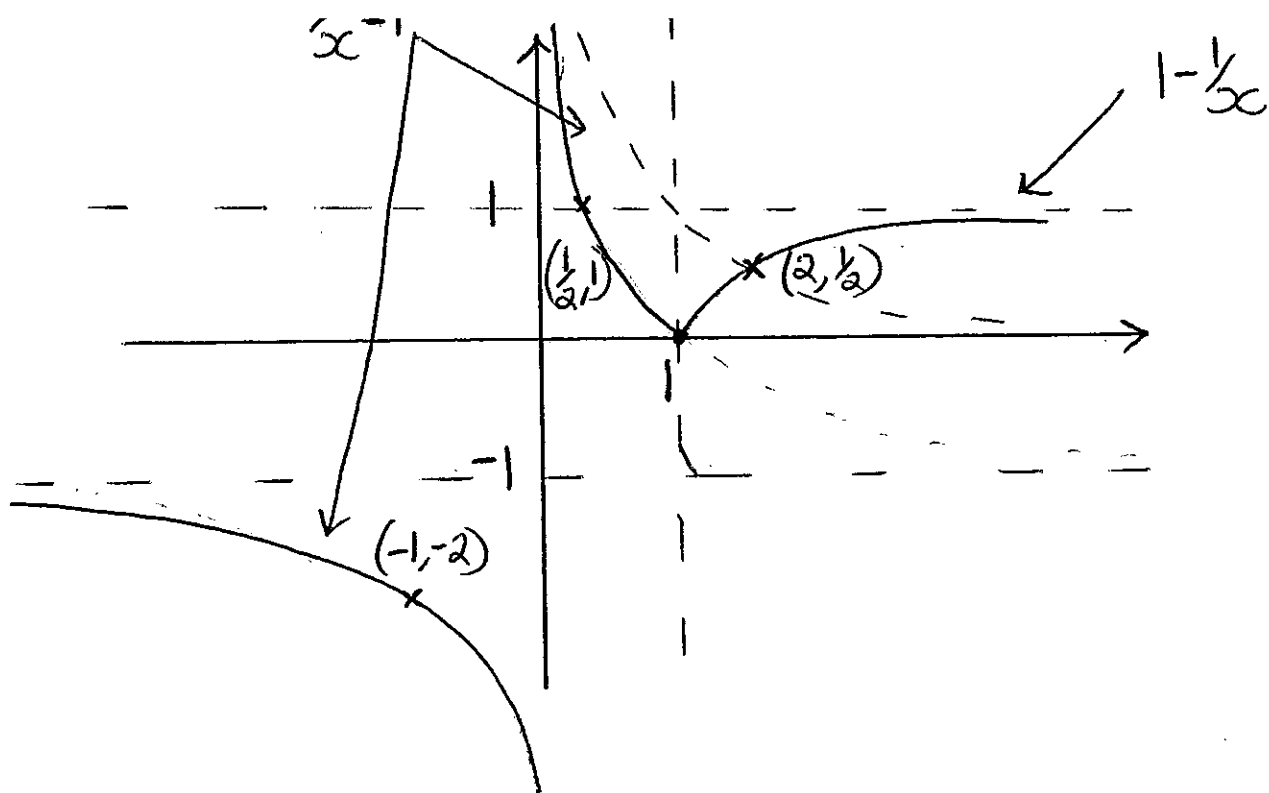


For $x > 1$ $y > 0$
 $-1 < x < 1$ $y < 0$
 $x < -1$ $y > 0$

b) $y = \frac{|x-1|}{x}$ Vertical asymptote $x = 0$ $\therefore$ No y intercept

Domain $x > 1$ $y = \frac{x-1}{x} = 1 - \frac{1}{x}$ as $x \rightarrow +\infty$ $y \rightarrow 1^-$
 $x < 1 (x \neq 0)$ $y = \frac{-(x-1)}{x} = \frac{1-x}{x} = \frac{1}{x} - 1$ as $x \rightarrow -\infty$ $y \rightarrow (-1)^-$

$y = 0 \Rightarrow x = 1$ x-intercept $(1, 0)$



Y4 Ext 1 Ass 1 2004

Question 1

a) i) $(x+2)(x^2-2x+4)$ ✓

ii) $5(x^2+x-12) = 5(x+4)(x-3)$ ✓

b) $\frac{1}{2+\sqrt{3}} \times \frac{2+\sqrt{3}}{2-\sqrt{3}} - \frac{1}{\sqrt{2}+1} \times \frac{\sqrt{2}-1}{\sqrt{2}-1}$ ✓

$= \frac{2-\sqrt{3}}{4-3} - \frac{\sqrt{2}-1}{2-1}$ ✓

$= 2-\sqrt{3} - \sqrt{2}+1 = 3-\sqrt{3}-\sqrt{2}$ ✓

c) $x = \frac{+4 \pm \sqrt{(-4)^2 - 4 \times 2 \times -7}}{2 \times 2}$ ✓

$= \frac{+4 \pm \sqrt{16 + 56}}{4}$

$= \frac{+4 \pm \sqrt{72}}{4}$

$= \frac{+4 \pm 6\sqrt{2}}{4} = \frac{+2 \pm 3\sqrt{2}}{2}$ ✓

d) $|2x-1| > 5$

$\pm(2x-1) > 5$

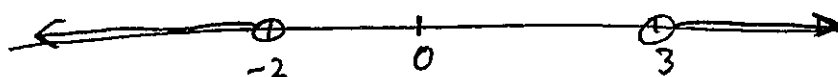
$2x-1 > 5$ $-(2x-1) > 5$

$2x > 6$ $2x-1 < -5$

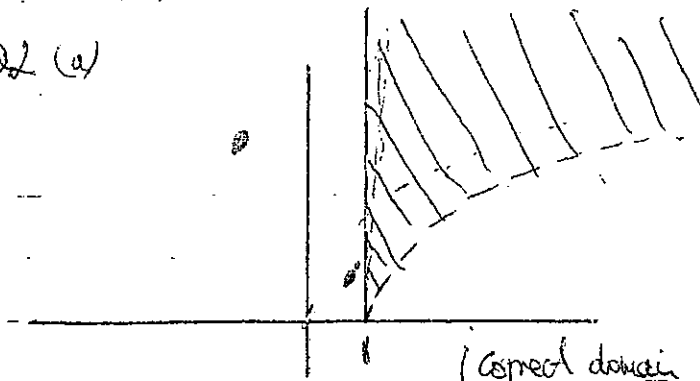
$x > 3$ $2x < -4$

$x < -2$ ✓

$x > 3$ or $x < -2$ ✓



Q2 (a)



1 correct domain
1 correct graph
1 broken line

(b) $(x \neq 0, y \neq 0)$

$$\frac{x-2}{y} + \frac{y+4}{x} = 0 \quad (x, y)$$

$$x(x-2) + y(y+4) = 0 \quad \checkmark$$

$$x^2 - 2x + y^2 + 4y = 0$$

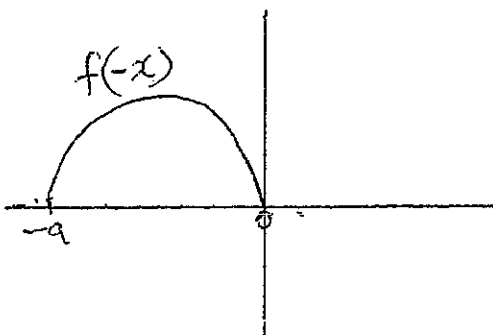
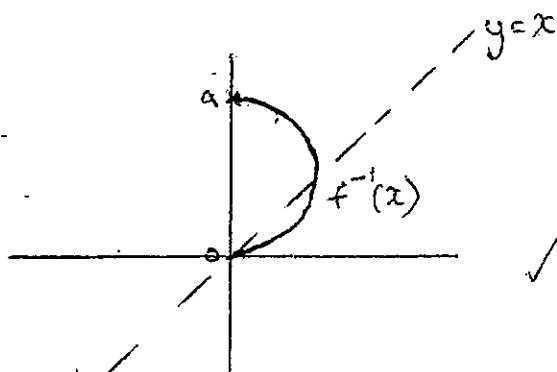
(complete the square for x and y)

$$x^2 - 2x + 1 + y^2 + 4y + 4 = 5$$

$$(x-1)^2 + (y+2)^2 = (\sqrt{5})^2 \quad \checkmark$$

$\therefore$ this is a circle with centre $(1, -2)$ and radius $\sqrt{5}$. (3)

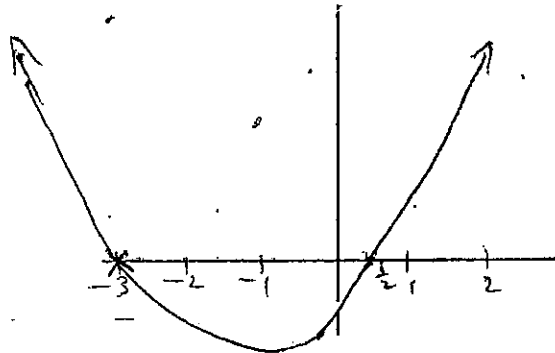
(c)



(2)

(d) $2x^2 + 5x - 3 \leq 0$

$$(2x-1)(x+3) \leq 0 \quad \checkmark$$

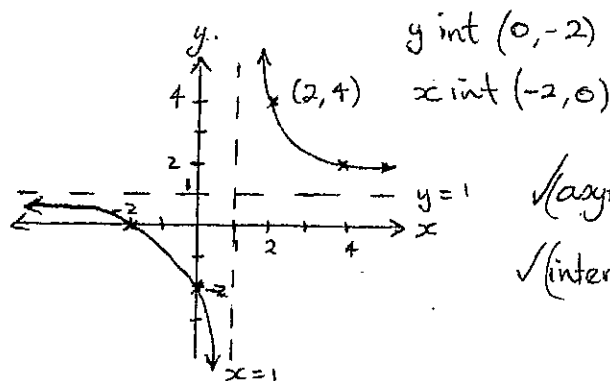


$$\therefore -3 \leq x \leq \frac{1}{2} \quad \checkmark \quad (2)$$

Question 3

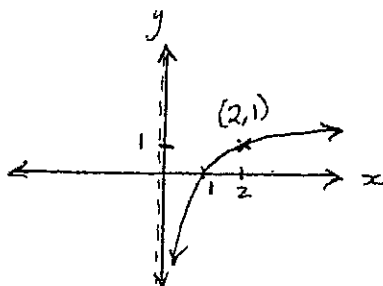
a) $\frac{x+2}{x-1} = \frac{x+1+3}{x-1}$
 $= 1 + \frac{3}{x-1}$ ✓ (working shown)

∴ hyperbola asympt: $y=1, x=1$



✓ (asympt.) 1mk Shape + one asymptote
 ✓ (intercepts) 2mk All asympt + intercepts

b) i/.



ii/ Function - since every relevant x -value produces only one y -value. ✓ (must have reason)
 - vertical line test holds.

iii/ Domain: $x > 0$ ✓

c) $\frac{4x}{x-1} \geq 1 \times (x-1)^2$ ✓ multiply by $(x-1)^2$

$4x(x-1) \geq (x-1)^2$

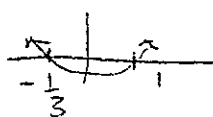
$4x(x-1) - (x-1)^2 \geq 0$

$(x-1)(4x - (x-1)) \geq 0$

$(x-1)(3x+1) \geq 0$

✓ quadratic inequality.

✓ solution of inequality.



∴ $x \leq -\frac{1}{3}$ or $x \geq 1$

But $x \neq 1$

∴ $x \leq -\frac{1}{3}$ or $x > 1$

✓ (restriction on domain)

NB: 1mk only given for correct solution of linear inequality (i.e. failure to multiply by square)

QUESTION FOUR

a) $y = \frac{1}{x^2-1} = \frac{1}{(x-1)(x+1)}$

(i) vertical asymptotes: $x = \pm 1$ ✓

(ii) $\lim_{x \rightarrow \pm \infty} \frac{1}{x^2-1} = 0$

$\therefore$ horizontal asymptote: $y = 0$ ✓

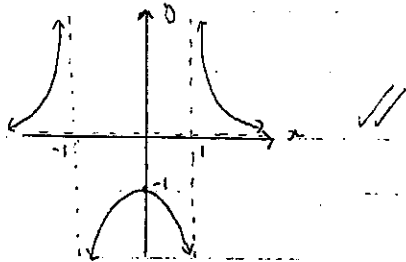
(iii) If $f(x) = \frac{1}{x^2-1}$ then

$f(-x) = \frac{1}{(-x)^2-1} = \frac{1}{x^2-1} = f(x)$ ✓

$\therefore$ function is even.

(iv) when $x = 0$, $y = -1$ } ✓
No x -intercepts.

(v)

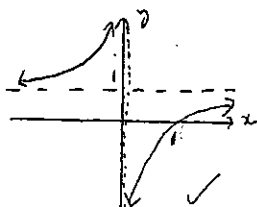


b) $|x-1| = x-1$ when $x \geq 1$

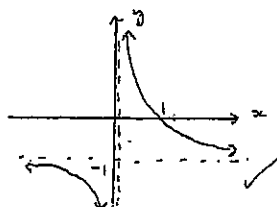
$|x-1| = 1-x$ when $x < 1$

when $x \geq 1$ $y = \frac{|x-1|}{x} = \frac{x-1}{x} = 1 - \frac{1}{x}$ } ✓
when $x < 1$ $y = \frac{|x-1|}{x} = \frac{1-x}{x} = \frac{1}{x} - 1$

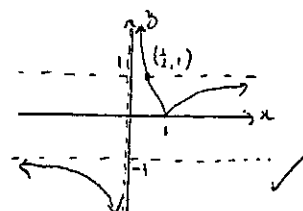
$y = 1 - \frac{1}{x}$



$y = \frac{1}{x} - 1$



$y = \frac{|x-1|}{x}$



Asymptotes are lines - they have equations. One mark deducted if both are not written as eqns. and a mark hasn't given been lost